

DR. ASIF ALI WAGAN

PhD, Computer Systems Engineering · M.E. Information Technology

Artificial Intelligence · Machine Learning · Bayesian & Probabilistic Modelling · Data Science · Clinical Decision Support

Computer Systems Engineering Department, Mehran University of Engineering & Technology, Jamshoro, Sindh, Pakistan (76062)

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PROFESSIONAL PROFILE

Doctorate-qualified computing researcher and university engineer with over twelve years of continuous academic service and a PhD in Computer Systems Engineering. Specialises in artificial intelligence, machine learning and probabilistic methods, with a doctoral focus on Bayesian and possible-world models for clustering uncertain, overlapping symptoms in clinical diagnosis. International research experience at the Universitat Politècnica de Catalunya (BarcelonaTECH), Spain, under a competitive HEC research-mobility grant. Founding In-Charge of a seven-node GPU high-performance-computing (AI-HPC) laboratory, with proven ability to commission, govern and administer advanced research infrastructure. A published author in indexed venues and an active graduate-research supervisor across undergraduate and Master's levels, with more than a decade of teaching across core computing subjects including Computer Architecture & Assembly Programming, Microprocessors & Interfacing, Artificial Intelligence, C++ Programming and Database Management Systems. Fluent in English (the medium of instruction throughout career), committed to high-quality undergraduate teaching, close student mentorship and rigorous, reproducible research.

EDUCATION

Doctor of Philosophy (PhD), Computer Systems Engineering

Feb 2018 – Jan 2026

Mehran University of Engineering & Technology, Jamshoro — CGPA 3.75 / 4.00

- **Thesis:** "Clustering Uncertain Overlapping Symptoms of Multiple Diseases in Clinical Diagnosis" — a bioinformatics and health-informatics study applying probabilistic and machine-learning methods to disease-symptom data for clinical decision support.
- Degree and transcript attested and recognised by the Higher Education Commission (HEC), Government of Pakistan.
- Comprehensive examination and thesis: Qualified. Coursework GPA 3.88 in the second semester (Internet of Things, Big Data Analytics, Advanced Image Processing, Data Mining Algorithms).

Master of Engineering (M.E.), Information Technology

Nov 2010 – Feb 2015

Mehran University of Engineering & Technology, Jamshoro — CGPA 3.80 / 4.00

- **Thesis:** "Uncertain Data Management using Probabilistic Database."

Bachelor of Engineering (B.E.), Computer Systems Engineering

Jan 2007 – Apr 2011

Mehran University of Engineering & Technology, Jamshoro — First Division

- Secured First Division across the full degree programme.

INTERNATIONAL RESEARCH TRIP (HEC-FUNDED)

Visiting Doctoral Researcher — CRAAX Lab

19 Mar 2022 – 20 Sep 2022

Universitat Politècnica de Catalunya (UPC BarcelonaTECH), Vilanova i la Geltrú, Spain

- Awarded the competitive **International Research Support Initiative Programme (IRSIP)** grant by the HEC to conduct a six-month doctoral research stay abroad (March–September 2022).
- Designed and validated a clustering algorithm based on the possible-world model using pre-computed probability values; validated on the TPC-H decision-support benchmark with promising results.

- Extended the work into federated and transfer-federated learning for distributed clinical-symptom data, implemented in Python (NumPy, Pandas, scikit-learn, Keras, PyTorch, Flower).
- Evaluated by the host supervisor as having produced “*excellent research work*” at the CRAAX Lab, UPC.

ACADEMIC & PROFESSIONAL EMPLOYMENT

Lab Engineer (BPS-18) — In-Charge, AI-HPC Lab

Dec 2024 – Present

Computer Systems Engineering Department, MUET, Jamshoro

- Founding In-Charge of the Department's **AI-HPC Lab** — a seven-node GPU workstation cluster (HP Z4/Z6 G5, Intel Xeon w5/w7, NVIDIA RTX 4070 Ti) for artificial-intelligence and machine-learning research and teaching.
- Led commissioning and hardware verification of the cluster against the donor handing-over record using Linux system diagnostics; authored the equipment-inventory and verification report adopted as the official departmental record.
- Authored the Lab's access-management policy and application framework — eligibility criteria, a tiered priority model and a weighted evaluation scheme governing student access to the high-performance resources.
- Administer the Ubuntu/Linux compute environment (CUDA, Python AI stack) and mentor undergraduate and Master's students on AI, data-science and systems projects; support laboratory delivery of core CSE courses.

Lab Engineer (BPS-17)

Jan 2019 – Dec 2024

Computer Systems Engineering Department, MUET, Jamshoro

- Managed laboratory instruction, equipment and student practical work across multiple core engineering courses while completing doctoral research.
- Supervised final-year and graduate project teams; maintained research computing resources for the department.

Lab Supervisor (BPS-16)

Feb 2013 – Jan 2019

Computer Systems Engineering Department, MUET, Jamshoro

- Delivered hands-on laboratory teaching and assessment support; administered laboratory infrastructure, networks and software environments for undergraduate cohorts.

TEACHING EXPERIENCE

Delivered laboratory and course instruction for undergraduate Computer Systems Engineering cohorts continuously since 2013, across the following subjects:

- Computer Architecture & Assembly Programming
- Microprocessors & Interfacing Techniques
- Artificial Intelligence
- Introduction to Computing & C++ Programming
- Database Management Systems

Responsibilities span lecture and laboratory delivery, design of practical exercises and assessments, project supervision and one-to-one student mentorship, reflecting a sustained commitment to quality teaching and close student interaction.

RESEARCH INTERESTS

Bioinformatics and Health Informatics · Clinical Decision Support · Artificial Intelligence and Machine Learning · Bayesian Networks and Probabilistic Graphical Models · Approximate and Exact Inference · Probabilistic & Uncertain Databases · Clustering of Uncertain Data · Federated and Distributed Learning · Graph Neural Networks · Data Mining and Big-Data Analytics.

PUBLICATIONS & MANUSCRIPTS UNDER REVIEW

Peer-Reviewed Journal

- A. A. Wagan, S. Talpur, and S. Narejo, “Clustering uncertain overlapping symptoms of multiple diseases in clinical diagnosis,” *PeerJ Computer Science*, 2024.

Peer-Reviewed Conference

- A. A. Wagan, S. Talpur, and S. Narejo, “Detecting Faulty Sensors by Analyzing the Uncertain Data Using Probabilistic Database,” 3rd Int. Conf. on Computing, Mathematics and Engineering Technologies (iCoMET), pp. 1–7, IEEE, 2020.

Manuscripts Under Review

- “Comparative Analysis of Approximate Inference Methods in Bayesian Networks Using Dirichlet-BDeu Parameter Estimation: A Clinical Diagnosis Case Study,” *SN Computer Science* (Springer Nature) — under review (Editor Assigned).
- “Probabilistic Inference of COVID-19 Diagnosis using Bayesian Network with Class-Imbalance Correction,” *Applied Computer Systems* (Riga Technical University) — under review.

RESEARCH SUPERVISION

- **M.S. (Cyber Security):** “Multi-Layer OSI Feature Fusion with Dual-Attention Graph Neural Networks for Continual Learning in Encrypted Mobile Traffic Classification” (in progress).
- **B.E. (Computer Systems Engineering):** “BugLens — An Error-Tracking Tool for Web Applications in Production” (completed, final-year project team).

TECHNICAL SKILLS

- **Languages & Frameworks:** Python (NumPy, Pandas, scikit-learn, Keras, PyTorch, Flower), Java, C/C++, PHP, .NET, HTML/JavaScript, Assembly.
- **Data & Databases:** PostgreSQL, MySQL, Oracle, MAYBMS (probabilistic database), data mining, big-data analytics.
- **Methods:** Bayesian networks, probabilistic inference, statistics & probability, modelling & simulation, machine learning, federated learning, clustering.
- **Systems & Tools:** Ubuntu/Linux server administration, virtualisation (VMware), Multisim, Keil, Arduino, embedded prototyping.

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

- Data Science & Big Data (IBM / Cognitive Class): Introduction to Data Science, Big Data 101, Hadoop 101, Data Privacy Fundamentals.
- Chromebook Data Science and Data Science workflow (Johns Hopkins University — Honors).
- System Administration Training (Linux server, Windows Server, virtualisation) and Software Development Training (PHP, MySQL, JavaScript, AJAX) — Hidayat Institute of Science & Technology.
- SAP Business Warehouse on SAP HANA and SAP Leonardo Machine Learning Foundation (openSAP); Google Online Marketing Fundamentals.

CONFERENCES & ACADEMIC SERVICE

- PhD Symposium Participant, 1st International Conference on Innovations in Information & Communication Technologies (IICT'26), MUET, 2026.
- Contributor, 3rd International Conference on Computing, Mathematics and Engineering Technologies (iCoMET 2020), Sukkur IBA University.
- Participant, 3rd International Multi-Topic Conference (IMTIC'13) including the Tutorial on Parallel Processing on GPUs, MUET.
- Organiser, Student Project Exhibition on Engineering Design (SPEED'16), ACM MUET Chapter, Department of Computer Systems Engineering.

LANGUAGES

English (full professional proficiency — medium of instruction throughout career), Urdu (native), Sindhi (native).

REFEREES

- **Dr. Sanam Narejo** — Chairperson, Computer Systems Engineering Department, Mehran University of Engineering & Technology, Jamshoro 76062, Sindh, Pakistan (present employer; PhD Co-Supervisor).
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Additional referees and full documentation available on request.